

Asian Autos

Chinese Autos: Why Automakers are entering Humanoid Robotics – and their competitive edge



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From Tesla and Hyundai, to XPeng and Xiaomi, an increasing number of automotive OEMs are entering the humanoid robotics space. In this report, we explore the strategic rationale, assess their competitive advantages, and highlight key investment implications.

Auto OEMs' push into humanoid robotics is recent but accelerating. Auto OEMs began moving into humanoid robotics from 2020. Hyundai announced its acquisition of 80% stake in Boston Dynamics in Dec 2020. Tesla then unveiled its "Tesla Bot" concept in August 2021. Since then, a growing number of Chinese OEMs have entered the field through: in-house development (e.g. XPeng and Xiaomi), subsidiary stakes (e.g. BYD's investments in PaXini and Zhiyuan), and strategic partnerships (e.g. Geely with UBTech). At the 2026 Beijing Auto Show, multiple OEMs (e.g. XPeng, Chery, BYD, etc.) showcased their humanoids, highlighting growing momentum in their broader tech strategies.

OEMs are entering humanoid robotics to boost productivity and unlock new revenue streams. Humanoids can further increase factory automation and drive structural cost reductions amid rising labor costs and workforce constraints. Applications may expand to consumer and service use cases (e.g., security, retail, home assistance), with some OEMs (e.g., Tesla, XPeng) viewing the TAM as comparable to or larger than autos over time.

Automakers have several advantages across hardware, software, and scale. There is significant overlap between vehicle and humanoid components—motors, reducers, sensors—as well as manufacturing. On the software side, there is partial overlap across robotics and autonomous driving in the development of VLA models and world models. With factory automation likely the first deployment wave, OEMs can pilot humanoids in-house, iterate rapidly, and scale deployments, accelerating learning while driving down unit costs. Strong balance sheets and R&D budgets further support long development cycles and talent attraction, positioning OEMs competitively in humanoid robotics.

Investment implications. Investors can gain exposure to humanoids **1) via OEMs or integrators** through dedicated players (e.g. UBTech, Unitree), automakers, or tech firms. Within Chinese autos, **XPeng** (Market-Perform) and **Xiaomi** (Outperform) stand out. XPeng's humanoid features a highly human-like design and explicitly targets consumer and household uses, potentially enabling higher margins if the company successfully addresses emotional companionship. XPeng's leadership in autonomous driving and broader physical AI reinforces its credentials in this space. We maintain our Market-Perform rating given the long timeline, but see strong optionality. Xiaomi is another notable player, and we see humanoid as a natural extension of its "Human × Car × Home" ecosystem as AI agents evolve. Another approach is **2) via the supply chain**. We prefer upstream components companies with exposure to broad robotics (not limited to the humanoid), broad customer base, proven success in expertise expansion, and high-quality core businesses, e.g. **Shuanghuan**, **Hesai**, and **Tuopu** ([here](#)). **We view ROI for humanoid robots in auto manufacturing** as compelling once payback falls below five years, vs. our current estimate of 7-8 years. Adoption should accelerate over the next 3-5 years, driven by advances in robotic intelligence, declining robot costs, and rising labor expenses.

BERNSTEIN TICKER TABLE

Ticker	Rating	22 Jun 2026		Price Target	TTM Rel. Perf.	Reported EPS				Reported P/E (x)		
		Cur	Closing Price			Cur	2025A	2026E	2027E	2025A	2026E	2027E
1211.HK (BYD)	O	HKD	78.35	136.00	(84.8)%	CNY	3.58	4.90	6.90	18.9	13.8	9.8
002594.CH (BYD)	O	CNY	87.59	124.00	(69.8)%	CNY	3.58	4.90	6.90	24.5	17.9	12.7
175.HK (Geely)	O	HKD	17.45	25.00	(39.2)%	CNY	1.67	1.82	2.26	9.0	8.3	6.7
2333.HK (Great Wall-H)	M	HKD	9.76	13.00	(63.5)%	CNY	1.16	1.21	1.40	7.3	7.0	6.0
2238.HK (GAC)	M	HKD	2.24	3.00	(66.2)%	CNY	(0.85)	(0.13)	(0.01)	(2.3)	(14.9)	(304.0)
LI (Li Auto)	M	USD	13.21	15.50	(74.9)%	CNY	1.13	(4.88)	2.27	79.1	(18.3)	39.4
2015.HK (Li Auto)	M	HKD	50.50	61.00	(97.5)%	CNY	0.57	(2.44)	1.13	77.1	(17.9)	38.4
9866.HK (NIO)	M	HKD	39.22	47.00	(0.4)%	CNY	(6.85)	(1.62)	(0.49)	(4.9)	(20.9)	(68.9)
NIO (NIO)	M	USD	5.02	6.00	21.5%	CNY	(6.85)	(1.62)	(0.49)	(5.0)	(21.0)	(69.1)
600104.CH (SAIC)	M	CNY	10.50	15.00	(79.4)%	CNY	0.89	0.97	1.05	11.9	10.8	10.0
1810.HK (Xiaomi)	O	HKD	23.72	43.00	(81.8)%	CNY	1.62	1.02	1.77	12.6	20.1	11.6
9868.HK (XPeng)	M	HKD	51.55	78.00	(75.9)%	CNY	(0.60)	(1.02)	0.23	(74.4)	(43.8)	197.2
XPEV (XPeng)	M	USD	13.21	20.00	(53.5)%	CNY	(1.20)	(2.03)	0.45	(74.7)	(44.0)	198.0
601689.CH (Tuopu)	O	CNY	60.19	75.00	(14.3)%	CNY	1.61	1.81	2.36	37.4	33.3	25.6
2050.HK (Sanhua)	M	HKD	26.70	27.00	(28.5)%	CNY	1.03	0.97	1.11	22.4	23.7	20.8
002050.CH (Sanhua)	M	CNY	45.23	39.00	34.3%	CNY	1.03	0.97	1.11	43.9	46.5	40.9
002472.CH (Shuanghuan)	O	CNY	42.11	60.00	(7.9)%	CNY	1.50	1.70	2.00	28.1	24.8	21.1
HSAL.US (Hesai)	O	USD	16.89	30.00	(53.8)%	CNY	2.92	3.31	5.27	39.1	34.5	21.7
2525.HK (Hesai)	O	HKD	129.90	238.00	NA	CNY	2.92	3.31	5.27	38.4	33.9	21.3
688017.CH (Leader Drive)	U	CNY	393.31	115.00	191.1%	CNY	0.69	0.83	0.96	568.0	471.5	409.0
300124.CH (Inovance)	O	CNY	68.95	82.00	(35.6)%	CNY	1.87	2.19	2.65	36.9	31.4	26.1
6324.JP (HDSI)	O	JPY	8,710.00	7,800.00	164.3%	JPY	16.99	57.37	79.51	512.7	151.8	109.5
ASIA			2,048.07									
SPX			7,500.58									
JPL			2,676.74									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

Source: Bloomberg, Bernstein estimates and analysis.

INVESTMENT IMPLICATIONS

Chinese Autos: For our EV names, we rate BYD and Xiaomi Outperform, and XPeng, Li Auto, and NIO Market-Perform. Within our traditional Chinese OEMs coverage, we rate Geely Outperform and Great Wall, GAC, and SAIC Market-Perform.

Asian Industrial Technology: We rate Inovance and Harmonic Drive Outperform.

Asia Emerging Robotics: We rate Shuanghuan, Hesai and Tuopu Outperform, Sanhua Market-Perform and Leader Drive Underperform

DETAILS

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AUTO OEM'S HUMANOID INITIATIVES AND PROGRESS

AUTO OEMS' PUSH INTO HUMANOID ROBOTICS IS RECENT BUT ACCELERATING

Auto OEMs began moving into humanoid robotics from 2020. Hyundai announced its acquisition of an approximately 80% stake in Boston Dynamics in December 2020, signaling its intent to build a broader robotics platform beyond vehicles. Tesla then unveiled its “Tesla Bot” concept (later named Optimus) at its AI Day in August 2021, putting general-purpose humanoids firmly on the automotive agenda.

Since then, a growing number of Chinese OEMs have entered the field through different routes: **1) in-house development** (for example XPeng's IRON robot and Xiaomi's humanoid projects), **2) subsidiary stakes** (such as BYD's investments in PaXini and Zhiyuan), and **3) strategic partnerships** (for instance Geely–Zeekr's collaboration with UBTech). This momentum has become increasingly visible at major industry events. At the 2026 Beijing Auto Show, multiple OEM-linked humanoids, including XPeng's IRON, Chery's AiMOGA, and BYD-related Zhiyuan AgiBot, were showcased publicly, underscoring how rapidly humanoid robotics is being woven into Chinese automakers' broader technology narratives.

INDUSTRY FRONTRUNNER WITH PLANS FOR NEARER-TERM COMMERCIALIZATION

- **Tesla** is developing its humanoid robot Optimus, progressing from Gen 1 (2022) to Gen 2 and Gen 2.5 prototypes by 2025, reflecting rapid iteration in hardware and software. Its strategy starts with manufacturing applications, with a long-term ambition to expand into consumer and household scenarios. Tesla targets limited commercialization in 2026 and volume shipments in 2027. A key constraint is that dexterous hand capability remains a major bottleneck, limiting real-world deployment readiness despite strong system-level progress.
- **Hyundai**, the parent company of Boston Dynamics, is pursuing an aggressive humanoid roadmap, transitioning Atlas from R&D to industrial deployment. Production-ready Atlas robots are being introduced into real factory environments, with initial applications in parts sequencing and heavy-duty manufacturing tasks. The group is targeting annual production capacity of up to 30,000 units by 2028, alongside internal rollout of over 25,000 robots across Hyundai facilities. This combination of full-

stack control, large-scale manufacturing plans, and clear volume targets positions Hyundai as the leading OEM in humanoid robot industrialization.

- **XPeng** is one of the more ambitious OEMs in humanoid robotics, with its IRON robot evolving through multiple generations during 2024-2025. A key milestone was its 2025 AI Day debut, where IRON's natural, catwalk-like walk went viral—so lifelike that audience questioned whether a human was inside. This showcased a major breakthrough in human-like locomotion and established XPeng as a frontrunner in embodied intelligence. The company targets mass production by end-2026 and global deliveries in 2027, focusing on both industrial and retail/service use cases such as showroom assistants and patrol robots, aiming for near-term commercialization.
- **Chery** is currently one of the more advanced OEMs in China on commercialization, with its humanoid robot “Moyin” achieving global delivery of 220 units in 2025 and further deployments across public service scenarios such as policing and medical guidance. Chery's humanoid robot are available for purchase for RMB 285.8k (US\$41k) through e-commerce channels like JD.com ([LINK](#)). Chery stands out for having the first meaningful batch delivery among OEMs, a diversified product ecosystem (including robot dogs and service robots), and a clear three-stage roadmap from companion robots to public service and eventually household applications.
- **GAC** has developed the GoMate humanoid series (now at the 4th-generation GoMate Mini), targeting applications in elderly care, security, and industrial environments, with pilot production planned for 2026 and mass production in 2027. Incrementally, GAC differentiates itself through innovations such as a wheel-legged hybrid mobility structure and by spinning off a dedicated robotics subsidiary to accelerate commercialization in a more market-oriented structure.

EARLY INDUSTRIAL DEPLOYER WITH REAL FACTORY VALIDATION

- **BMW** has rapidly progressed humanoid robotics from pilot testing to real production environments, building on early collaborations with Figure's robots in 2025. At its Spartanburg plant, humanoids supported the production of over 30k vehicles through tasks such as sheet-metal handling, demonstrating reliability in high-throughput settings. The company is now expanding pilots to Europe, with deployments in Leipzig targeting battery assembly, intralogistics, and component production from summer 2026. BMW's strategy emphasizes iterative scaling through live manufacturing validation, positioning humanoids as flexible co-workers rather than committing to immediate mass production
- **Toyota** is among the first OEMs to convert humanoid pilots into commercial deployment through a Robots-as-a-Service (RaaS) model with Agility Robotics. Following a successful pilot, Toyota signed a 2026 agreement to deploy Digit humanoids in production, focusing on logistics tasks such as parts handling and line feeding. Initial deployments remain small-scale, but Toyota's asset-light RaaS approach enables flexible scaling while reducing upfront costs and technology risk.

EMERGING PLAYERS WITH STRONG POTENTIAL IN THE LONG TERM

- **Xiaomi** has been developing humanoid robots since 2020, launching CyberOne in 2022 and more recently open-sourcing its Xiaomi-Robotics-0 embodied AI model in 2026. Its current focus is on manufacturing scenarios such as inspection and assembly, though no clear mass production timeline has been announced. Xiaomi has demonstrated strong technical progress, including achieving over 90% success rates in real factory tasks and advancing high-precision dexterous hand capabilities, supported by its strength in AI foundation models and embodied intelligence.
- **BYD** is advancing an internally developed humanoid robot project (codename “Yao Shun Yu”), initiated in 2022 and supported by partnerships such as its embodied intelligence lab with HKUST. BYD stands out for its deep vertical integration across batteries, motors, semiconductors, and precision manufacturing, as well as its potential to leverage its global dealership network for future commercialization.
- **Li Auto** is taking a differentiated approach by framing robotics under a broader “space robot” concept, incorporating wheeled robots for manufacturing and future humanoids potentially for household use. While mass production plans are not disclosed, the company has established dedicated robotics business units. Li Auto is notable for its emphasis on AI, including heavy investment in large models such as Mind GPT, and its vision of integrating robots into a wider in-car, wearable, and intelligent ecosystem.

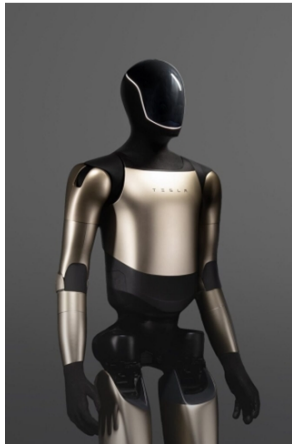
EXHIBIT 1: **Overview of Auto OEMs progresses in humanoid robots**

OEM	Platform	Timeline	Progress & Goals	
			Application	Shipment
Tesla	Optimus (In-house)	Aug 2021: Unveiled conceptual design at AI Day Sep 2022: Gen 1 prototype was shown Sep 2023: Video showing Optimus sorting coloured blocks Dec 2023: Gen 2 was showcased May 2024: Footage of Optimus performing tasks in a Tesla factory	- Deploy Optimus inside Tesla factories for "boring, repetitive and dangerous" tasks - Longer term, commercialize for industrial and domestic use	- Targeting "thousands of bots" in Tesla factories and initial external shipment from 2026; Gen 3 production will start around summer 2026, with high-volume ramp targeted for 2027 - Fremont production line is designed for 1mn robots/year and Giga Texas planned for 10m/year from 2027
BMW	Figure AI - Figure 02/03 (Partnership)	Jan 2024: BMW announced collaboration with Figure AI	- Integrate humanoids into body shop at Leipzig / Spartanburg plants for "monotonous, ergonomically demanding, or safety-critical tasks"	Figure AI's Gen-2 humanoid robot has participated in the production of 30,000 BMW cars
Hyundai	Boston Dynamics - Atlas/ Spot (Subsidiary, 80% stake)	- 2021: Acquisition of Boston Dynamics - 2026: Launch of new Atlas ready for volume shipment	- Internal use first; start with parts sequencing tasks and expand to component assembly later	- By YE 2025: 2k units of Spot deployed - By 2028: Aim to build manufacturing capacity for 30,000 Atlas units annually
Toyota	Agility Robotics - Digit (Partnership)	Oct 2024: Announced research partnership with Boston Dynamics Feb 2026: Deploy Agility Robotics' robots in Canada factory	- Use Agility Robotics' Digit to build newest RAV4 in Toyota Canada	N.A.
BYD	Yao Shun Yu (In-house) / AgiBot (Minority stake)	2023: Invested in AgiBot (Zhiyuan) 2025: Confirmed they are developing humanoid robotics	Focus is on deployment in factories to increase automation	N.A.
XPeng	IRON (In-house)	Nov 2024: Unveiled IRON robot; Nov 2025: Released Gen 2 IRON	- Prioritizing commercial retail / sales scenarios i.e. showroom guidance, retail assistance, patrol services; also exploring industrial applications and domestic use case	- Began construction of a mass production facility in Guangzhou in Q1 2026, and targeting mass production (c.1k units) by year-end 2026 - Plan for the humanoid robots to station at XPeng retail stores as shopping guides
Xiaomi	CyberOne (In-house)	Aug 2022: Debuted CyberOne Feb 2026: Open-sourced Xiaomi-Robotics-0 (4.7B-parameter embodied VLA model); Continued R&D and demonstration	- Initially in Xiaomi's own manufacturing plants, tasked with repetitive assembly line duties - Longer term, it will be commercialized for domestic use case, and be integrated into the smartphone, AIOT, and EV ecosystem	- Began pilot deployment of humanoid robots in its EV manufacturing plant in Beijing - Expect humanoid robots will operate on a "large-scale" in Xiaomi's factories within 5 years
GAC	GoMate (In-house)	2022: Began R&D Dec 2024: Released Gen 3 humanoid GoMate Feb 2026: Launched Gen 4 GoMate Mini	Pilot deployments in elderly care and security	Targeting small-scale pilot production in 2026 and mass production in 2027
Geely	UBTech - Walker S Lite (Partnership)	Jul 2024: Deployed at Zeekr factory	- Primarily in auto manufacturing scenarios, including material handling and logistics, assembling, sorting and line-side tasks, as well as quality inspection	Still in early pilot stage till now
Chery	Aimoga (In-house)	Established robotics company in Jan 2025	- Robots: currently for smart policing and medical guidance; plan to expand to household scenarios in longer term - Robotic dogs: home companionship, community patrol, and industrial site security	Have delivered hundreds of robots with 1k+ order in hand
Li Auto	N.A.	Established robotics team in 2025	Two-wheeled robots are primarily used in factory manufacturing, while bipedal robots are developed for household scenarios	N.A.

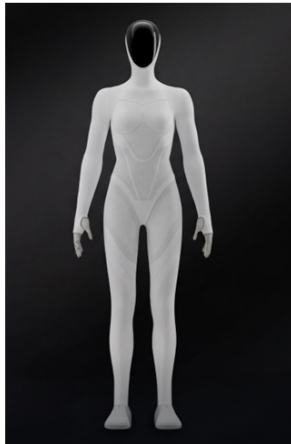
Source: Company websites, Forbes, 36Kr and Bernstein Analysis

EXHIBIT 2: Overview of Auto OEMs' humanoid robot products

Tesla Optimus v2.5



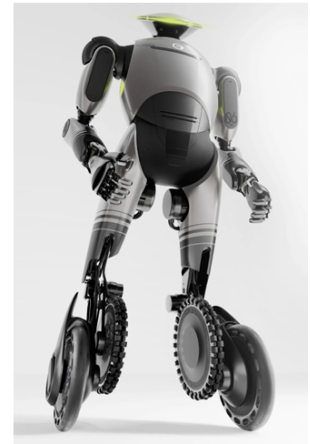
XPeng IRON



Xiaomi CyberOne



GAC GoMate



Source: Company websites and Bernstein analysis

EXHIBIT 3: Chery's Aimoga



Source: Bernstein photo

EXHIBIT 4: AgiBot was showcased at BYD's exhibition



Source: Bernstein photo

WHY AUTO OEMS ARE EXPANDING INTO HUMANOID ROBOTS?

Auto OEMs are expanding into humanoid robotics for two main reasons: to raise internal productivity and to open up new revenue pools beyond the core vehicle business. They also believe they possess structural advantages in manufacturing, supply chains, and embodied AI that position them well in this emerging category.

On raising internal productivity: Humanoid robots offer a logical next step in factory and warehouse automation, especially as manufacturers face rising labour costs, an aging workforce, and persistent shortages in repetitive, physically demanding, or harsh-environment roles. While stamping, welding, and painting are already highly automated, final assembly and intralogistics remain comparatively labour-intensive. This leaves a meaningful automation gap in tasks such as material handling, precision assembly, inspection, and testing. Humanoid robots could help narrow that gap by operating in tighter spaces and more complex shop-floor environments than traditional fixed automation. Material handling is a particularly relevant use case, given its high injury incidence and recurring labour shortages during peak production periods. If execution improves and costs fall, humanoids could support both labour substitution and structurally lower manufacturing costs over time.

Opening up new external revenue streams: Some OEMs, including Tesla and XPeng, have framed the long-term total addressable market for humanoid robots as comparable to, or potentially larger than, the automotive market. In addition to manufacturing and warehouse settings, humanoids could eventually address a broad range of consumer and service applications, including patrol and security, retail guide and store operations, and, over the longer term, household assistance. For OEMs, the appeal is not only participation in a potentially large new market, but also the opportunity to extend their capabilities in high-volume manufacturing, supply chain know how, software, sensing, and control systems into a new product category.

EXHIBIT 5: Why Auto OEMs are investing in humanoid robots

Rationale	Applications	Details
Internal productivity boost	Factory and warehouse automation, including material handling, assembly, quality inspection, etc.	Humanoids can further increase factory automation and drive structural cost reductions. They help address rising labor costs, aging workforces, and shortages in repetitive, physically demanding, or harsh-environment roles.
External revenue streams	Range of consumer and service applications, including patrol & guard, retail guide, household helper, etc.	OEMs believe they have an advantage in humanoids, and want to take part in the potential multi-trillion-dollar market opportunity over the long term (2040+).

Source: Bernstein analysis

APPLICATION SCENARIOS

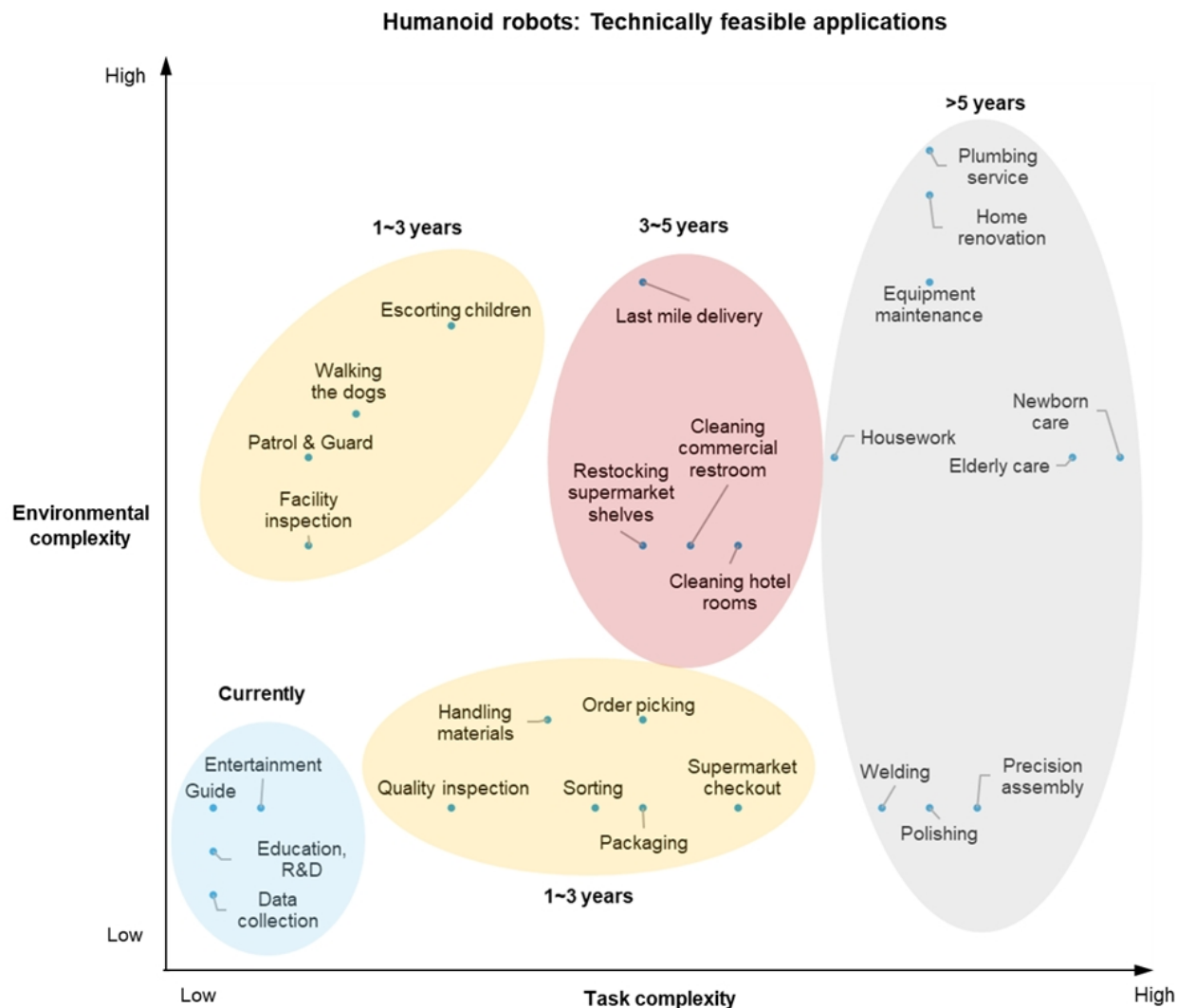
Manufacturing is one of the most clearly defined application for humanoid robots by automotive OEMs at this stage, given the potential ROI from replacing human labor. Auto OEMs are using their own factories for initial pilot and deployment, to validate technology and drive down costs. BYD confirmed their expansion into humanoid robotics in June 2026. Xiaomi's humanoid pilot achieved over 90% success rates in assembly tasks. ([Link to video: Inside Xiaomi's EV Factory : Xiaomi's Robot Breakthrough Revealed](#).) GAC and Seres are moving from pilot to early scaling. Overall, manufacturing serves as the foundation, with a clearer path from trial deployment to large-scale rollout.

Consumer-facing and service applications, such as retail and showroom guide, patrol & security are emerging as the next phase, focused on customer-facing roles such as in-store assistants, brand interaction, and public services. This scenario places higher demands on perception, interaction, and mobility, making it less mature than manufacturing but critical for commercialization and brand differentiation. XPeng is leading this push, planning to deploy humanoid robots as in-store sales assistants from late 2026 and expand commercially by 2027. Chery has deployed over 200 robots in events and exhibitions for entertainment, and public service use cases. GAC and Changan are also exploring security and service applications, though still at early stages. To summarize, this segment is transitioning from pilot deployments to initial commercialization.

Household applications represent a longer-term opportunity for humanoids—still at an early stage, but with the largest potential market. Use cases such as home cleaning, elderly care, and butler or companionship services will require significant breakthroughs in general intelligence, safety, and cost. As a result, most OEMs view this as an endgame objective. Chery has outlined one of the clearest roadmaps, progressing from companion robots to service applications and eventually to in-home use, with early consumer-channel experiments already underway. Changan plans to enter the household segment after

achieving mass production around 2028. Xiaomi, meanwhile, positions humanoid robots as a natural extension of its “Human × Car × Home” ecosystem in the next stage of embodied AI development.

EXHIBIT 6: **Humanoid robots: Technically feasible applications across the near-, mid-, and long-term future**



Source: Bernstein analysis and estimates

EXHIBIT 7: **Where and in what applications can humanoid robots be used**

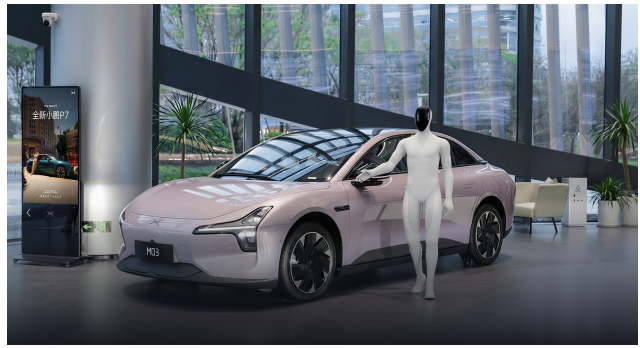
Source: Bernstein analysis and estimates

EXHIBIT 8: Geely began using humanoid robots in the production of Zeekr electric vehicles



Source: Shenzhen Daily

EXHIBIT 9: XPeng plans to deploy 1,000 robots at its showrooms by year-end 2026



Source: XPeng

EXHIBIT 10: Chery's Aimoga is used in public security scenarios



Source: Bernstein pictures

EXHIBIT 11: Hexagon's humanoid robot AEON at BMW's Leipzig plant



Source: BMW

AUTO OEMS' RIGHT TO WIN IN HUMANOID ROBOTICS

Automakers have several advantages that make them well positioned in humanoid robotics, given the significant overlap between the automotive and humanoid robotics hardware and software ecosystems, access to resource, and scale and cost

advantages.

- **Manufacturing and supply chain know-how.** On the hardware side, auto and humanoid manufacturing share substantial similarities, including component overlap in motors, batteries, reducers, and sensors such as LiDAR and cameras. Auto OEMs' expertise in highly automated, large-volume production helps them tackle key challenges in yield, consistency, quality, and reliability in humanoid manufacturing.
- **Physical AI leadership.** From a software perspective, we view that there is partial overlap across robotics and autonomous driving in the development of VLA models and world models. Humanoid robots rely on many of the same foundational technologies as autonomous vehicles, including end-to-end learning algorithms, world models, simulation systems, and multi-modal sensing fusion. A decade of investment in autonomous driving has given leading OEMs deep engineering expertise, strong R&D teams, large datasets, and robust compute infrastructure. Many OEMs therefore view humanoids as a natural next step in 'physical AI', where they can leverage their existing AV technology stack for humanoid development.
- **Scale and cost advantages.** With factory and warehouse automation as the first wave of humanoid deployment, auto plants provide an ideal testbed. OEMs can pilot humanoids in their own facilities, iterate quickly, and then scale deployments across global plant networks. Automakers' scale in procurement and their established supplier relationships enable them to source humanoid components at competitive costs. This ability to deploy at scale early should help drive down unit costs, improve reliability, and accelerate learning relative to many other industries.
- **Resources and talent access.** Compared with most humanoid start-ups, established auto OEMs benefit from stronger balance sheets, larger R&D budgets, and existing global brands. This gives them more capacity to fund long development cycles and makes them more attractive to top talent in AI and robotics.

While not guaranteeing success, these advantages in manufacturing, supply chains, and embodied AI position OEMs competitively in the emerging humanoid robotics market. Humanoid start-ups often bring innovation, while automotive OEMs are typically stronger at large-scale industrialization and commercialization. Although many start-ups are led by leading scientists who can define new technological directions, bringing humanoid robots to market is fundamentally an engineering and integration challenge. It requires solving practical issues such as reducing the gap between model-predicted actions and real-world behaviour, selecting materials that balance low weight with structural strength, designing joints with high torque density, and meeting stringent cost and energy-efficient constraints. These problems call for coordinated, cross-functional execution rather than isolated technical breakthroughs. In China's highly competitive environment, early laboratory-level leadership tends to be short-lived, whereas robust engineering capabilities and scalable manufacturing know-how are more likely to support durable competitive advantages.

EXHIBIT 12: **Auto OEMs have several structural advantages in humanoid robotics**

Advantages	Details
Manufacturing and supply chain know-how	- Auto and humanoid manufacturing share substantial similarities, including component overlap with EVs (e.g. motors, batteries, reducers, sensors, etc.) - Proven expertise in highly automated, large-volume production helps them tackle key challenges in yield, consistency, quality, and reliability in humanoid manufacturing
Physical AI leadership	- Leverage autonomous driving tech stack: end-to-end learning, world models, simulation, sensor fusion, etc. - Years of autonomous vehicle investment provide large datasets, robust computing infrastructure, sophisticated models, and experienced engineering teams
Scale and cost advantages	- Own factories provide ideal testbeds for pilots, iteration, and deployment - The ability to deploy at scale early should help drive down unit costs, improve reliability, and accelerate learning relative to many other industries - Automakers' scale in procurement and their established supplier relationships enable them to source humanoid components at competitive costs
Resources and talent access	- Strong balance sheets and R&D budgets to fund long development cycles - Access to resource and brand recognition helps attract top talent in AI and robotics

Source: Bernstein analysis

INVESTMENT IMPLICATIONS

For investors, we think there are two primary approaches to gaining exposure to humanoid development. **The first is investing at the OEM or robot integrator level**—whether dedicated humanoid players (e.g. UBTech, Unitree, etc.), automotive OEMs, or technology companies with humanoid ambitions. This route ultimately hinges on optionality versus capital intensity (capex and opex), with a risk-reward profile that can be relatively binary depending on execution and the broader competitive landscape.

Within our coverage, XPeng and Xiaomi stand out as relatively compelling among automotive OEMs. XPeng's humanoid robot, IRON, features a highly human-like design and is explicitly targeting consumer-facing and household use cases. In our view, this positioning could enable structurally higher margins if the company successfully addresses emotional companionship, leveraging both form factor and AI capabilities. XPeng's leadership in autonomous driving and broader physical AI further reinforces its credentials in this space. While we maintain a Market Perform rating given the long commercialization horizon, we recognize the company's strong optionality in humanoids.

Xiaomi is another notable player. The company has made rapid progress in AI, with developments such as the MiMo model and the MiClaw agent. We see humanoids as a natural extension of Xiaomi's "Human × Car × Home" ecosystem as the industry transitions toward AI agents. In this framework, humanoids could serve as the physical embodiment of AI—integrating smartphones, AIoT devices, and mobility. Xiaomi's extensive installed base of connected devices provides a potentially unparalleled distribution and data advantage. We rate Xiaomi Outperform. Xiaomi's stock price is currently depressed by high memory costs, but the company continues to progress in smartphone premiumization, has significant overseas potential in IoT and EVs, and made notable inroads in AI development.

The second investment approach is via the supply chain, including key components such as motors, harmonic reducers, vision products, sensors, and power and AI semiconductors. This pathway may offer more diversified exposure with potentially less binary outcomes, as adoption unfolds across multiple OEMs. We prefer upstream components companies with exposure to broad robotics (not limited to the humanoid), broad customer base, proven success in expertise expansion, and high-quality core businesses, e.g. **Shuanghuan** (Outperform), **Hesai** (Outperform), and **Tuopu** (Outperform) ([here](#)).

HUMANOID ROBOTS IN AUTO MANUFACTURING: WHEN DOES ROI BECOME COMPELLING?

In assessing the ROI implications of using humanoid robots to replace labor in automotive manufacturing, we believe the discussion becomes more compelling when the payback period falls below five years. Based on current assumptions, i.e., a humanoid robot cost of approximately US\$50,000 and an entry-level auto manufacturing worker in China earning around US\$13,000 annually, we estimate a payback period of roughly 7-8 years. We expect adoption to accelerate over the next 3-5 years however, driven by advances in robotic intelligence that improve task-specific capabilities. At the same time, we anticipate humanoid robot costs to decline below US\$40,000, while labor costs rise above US\$14,000, making the ROI dynamics increasingly attractive. ([Exhibit 13](#) - [Exhibit 15](#))

EXHIBIT 13: Payback analysis of humanoid robots in China's auto manufacturing plants

Estimated payback period of humanoid robots in China's auto manufacturing				
- Assume 1 humanoid robot to replace 0.75 entry-level manufacturing worker (i.e. 75% human worker efficiency)				
- Assume 1 human operator to look after 3 humanoid robots				
Illustration of economics (in USD)				
Upfront investment in humanoid robot solutions				
Upfront investment	# of humanoid robots	Robot unit price	Integration cost (% of robot price)	Sub-total
Procurement and system integration of humanoid robots	2	50,000	30%	130,000
Before replacement (entry-level manufacturing workers)				
Annual labor cost	# of workers	Unit labor cost	Implied shifts (relative to robot)	Sub-total
Manufacturing labor	1.5	12,857	2.6	69,796
After replacement (humanoid robots + human operator/ supervisor)				
Annual robot operating and maintenance cost	# of humanoid robots	Annual cost (% of robot price)	Unit cost	Sub-total
Humanoid robots	2	15%	7,500	15,000
Annual labor cost	# of operators	Unit labor cost	Implied shifts (relative to robot)	Sub-total
Operators	0.7	15,429	2.6	37,844
Annual cost saved				16,951
Payback period (years)				7.7

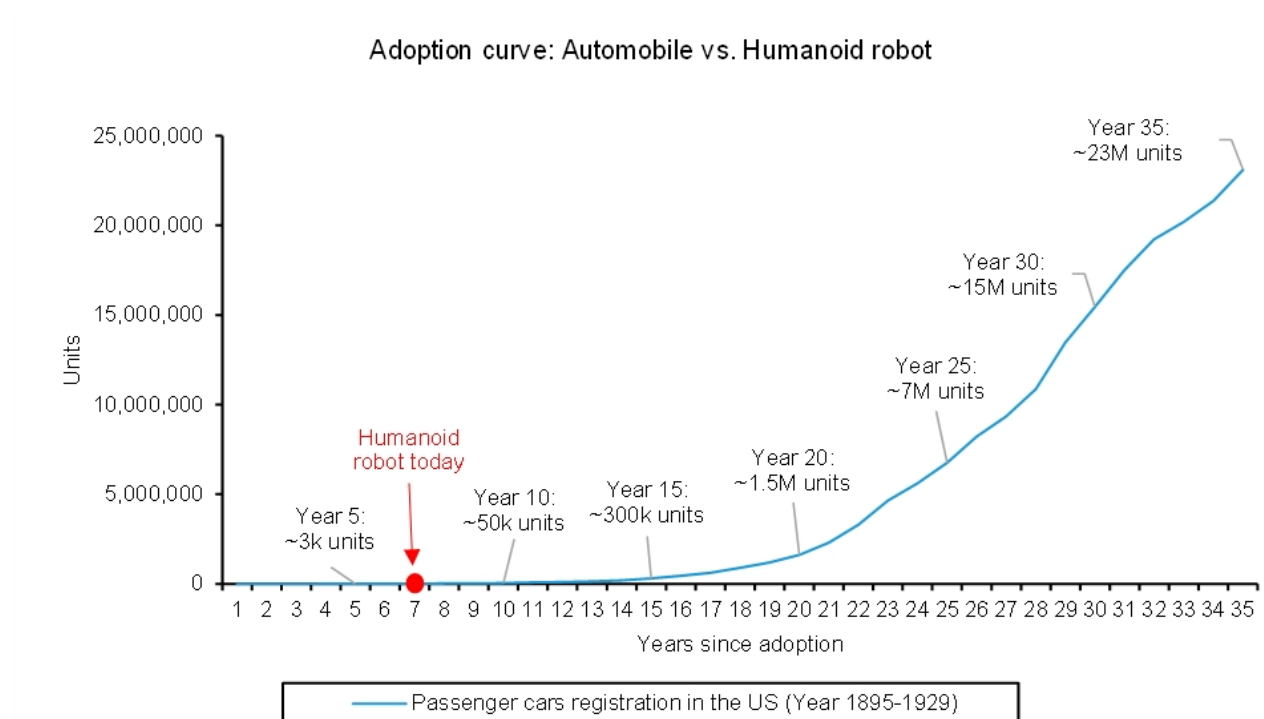
Assumptions	
Unit labor cost	
Basic annual wage (RMB 5k/ month for entry-level manufacturing labor; calculation in USD)	8,571
Plus: Housing, meals, welfare, and miscellaneous (% of basic annual wage)	50%
Total annual labor cost (entry-level manufacturing labor, in USD)	12,857
Assume human operator / robot supervisor costs 20% more (in USD)	15,429
Working time & shifts	
Working hours per day - human	10
Working hours per day - robots	22
Implied shifts per day	2.2
Working days per year - human (i.e. 49 weeks * 6 days per week)	294
Working days per year - robot (assume 15 days of maintenance per year)	350
Implied shifts per year (human relative to robot)	2.6

Source: Bernstein estimates and analysis

EXHIBIT 14: **Sensitivity analysis of the payback period to humanoid robot prices and labor cost**

Payback period (years)		Humanoid robot unit price (USD)									
		100,000	90,000	80,000	70,000	60,000	50,000	40,000	30,000	20,000	10,000
Unit labor cost (USD)	10,000	180.4	162.4	144.3	126.3	108.2	90.2	72.2	54.1	36.1	18.0
	11,000	37.8	34.1	30.3	26.5	22.7	18.9	15.1	11.4	7.6	3.8
	12,000	21.1	19.0	16.9	14.8	12.7	10.6	8.5	6.3	4.2	2.1
	13,000	14.7	13.2	11.7	10.3	8.8	7.3	5.9	4.4	2.9	1.5
	14,000	11.2	10.1	9.0	7.9	6.7	5.6	4.5	3.4	2.2	1.1
	15,000	9.1	8.2	7.3	6.4	5.5	4.5	3.6	2.7	1.8	0.9
	20,000	4.7	4.2	3.7	3.3	2.8	2.3	1.9	1.4	0.9	0.5
	25,000	3.1	2.8	2.5	2.2	1.9	1.6	1.3	0.9	0.6	0.3
	30,000	2.4	2.1	1.9	1.7	1.4	1.2	0.9	0.7	0.5	0.2
	40,000	1.6	1.4	1.3	1.1	0.9	0.8	0.6	0.5	0.3	0.2
	50,000	1.2	1.1	1.0	0.8	0.7	0.6	0.5	0.4	0.2	0.1

Source: Bernstein estimates and analysis

EXHIBIT 15: **Adoption speed of automobiles in the early years can be regarded as a reference**

Note: Registration is more like a concept of "install base", instead of "annual shipment".

Source: "Facts and Figures of the Automobile Industry 1920-1930", Bernstein analysis and estimates

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